25900240

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEBIF201 - LINUX AND C MASTERY

Duration: 1 Hour 30 Minutes Maximum Marks: 50

(2024 ADMISSION ONWARDS)

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any 5 questions

Each question carries 2 marks

1. Explain about the keywords in C. [K]/[CO1]

Keywords are reserved words in C used for predefined purposes. There are 32 keywords (e.g., int, if, return). They cannot be used as identifiers.
Mark distribution: Definition – 1 mark, Examples – 1 mark

2. Discuss the role of the kernel in process management. [U]/[CO3]

Kernel manages process creation, termination, context switching, and resource allocation.
Mark distribution: Explanation – 1 mark, Functions – 1 mark

3. Write the basic syntax for a while loop. [K]/[CO8]

Syntax:

```
while(condition){
  // statements
}
```

Mark distribution: Syntax – 1 mark, Explanation – 1 mark

4. Illustrate the use of break in a for loop. [A]/[CO2]

The break statement terminates loop execution prematurely.
Example:

```
for(i=0;i<10;i++){
 if(i==5) break;
}
```

Mark distribution: Concept – 1 mark, Example – 1 mark

5. Describe the use of the chmod command in Linux. [U]/[CO4]

chmod changes file permissions.
Syntax: chmod [options] mode file.
Example: chmod 755 file.
Mark distribution: Command purpose – 1 mark, Example – 1 mark

6. Explain the difference between scanf() and printf() functions. [U]/[CO1]

scanf() is used for input, printf() for output.
Example: scanf("%d", &x); printf("%d", x);
Mark distribution: Difference – 1 mark, Example – 1 mark

7. Create a pipeline that counts the number of lines in a file containing a specific word. [A]/[CO5]

Command: grep 'word' filename | wc -l
Mark distribution: Command – 1 mark, Explanation – 1 mark

[2 × 5 = 10]

Part B

Short Essay Questions

Answer any 4 questions

Each question carries 5 marks

8. What do exit status codes represent in shell scripting? Give examples. [K]/[CO8]

Exit codes indicate command success or failure.
0 = Success, Non-zero = Failure.
Example: exit 0; exit 1; $? checks last command.
Mark distribution: Definition – 1, Meaning – 2, Example – 2

9. Explain 1-D array with suitable example. [A]/[CO2]

A 1-D array stores multiple values of the same type.
Syntax: int arr[5] = {1,2,3,4,5};

Access: arr[i]
Mark distribution: Definition – 1, Syntax – 2, Example – 2

10. Describe how typedef improves readability in user-defined data types. [U]/[CO1]

typedef gives an alias name to existing data types.
Example: typedef unsigned int uint;
Improves code clarity.
Mark distribution: Concept – 2, Example – 2, Benefit – 1

11. Write the command to check the connectivity to google.com. [A]/[CO6]

Command: ping google.com
Displays ICMP echo replies.
Mark distribution: Command – 2, Explanation – 3

12. Explain the importance and method of making a shell script executable. [K]/[CO8]

Use chmod +x scriptname to make a script executable.
Purpose: To run directly with ./script.sh
Mark distribution: Command – 2, Explanation – 3

13. Explain the purpose of conditional statements in shell scripting with an example. [A]/[CO2]

Conditional statements control program flow.
Example:
if [$num -gt 0]; then
echo "Positive"
else
echo "Negative"
fi
Mark distribution: Explanation – 2, Syntax – 1, Example – 2

[5 × 4 = 20]

Part C

Long Essay Questions

Answer any 2 questions

Each question carries 10 marks

14. Apply an if-else statement to check whether a number is positive or negative. [U]/[CO1]

```
#include <stdio.h>
int main(){
```

```
 int num;
 printf("Enter number: ");
 scanf("%d", &num);
 if(num>0)
 printf("Positive");
 else if(num<0)
 printf("Negative");
 else
 printf("Zero");
 return 0;
}
```

Mark distribution: Syntax – 2, Logic – 4, Program output – 4

15. a) Describe two ways to debug a shell script effectively.
b) Explain the role of set -x and set +x in debugging. [A]/[CO6]

Debugging methods:
1. Using echo statements.
2. Running script with bash -x.
set -x enables debug tracing, set +x disables it.
Mark distribution: Explanation – 4, Example – 6

16. Explain why the documentation section is important in large C programs with an example. [K]/[CO8]

Documentation improves readability and maintainability.
Example:
/* Program: Addition
Author: Student
Date: 03/11/2025
Description: Adds two numbers */
Mark distribution: Definition – 3, Example – 4, Importance – 3

17. Explain how to change file permissions using chmod with an example. [A]/[CO2]

Command: chmod 755 file
Owner: rwx, Group: r-x, Others: r-x.
Mark distribution: Syntax – 2, Example – 4, Explanation – 4

[10 × 2 = 20]